

Reza Khanahmadloo

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with hands-on experience building scalable batch and streaming data pipelines using Python, SQL, Apache Spark, Apache Airflow, Kafka, Microsoft Fabric, and cloud data platforms. Passionate about designing scalable ETL/ELT workflows, data platforms, and modern analytics solutions through real-world projects. Enjoy collaborating with architects, software engineers, and DevOps teams.

CORE TECHNICAL SKILLS

Programming & Analytics Tools

- Python (Data Processing, Automation, Pandas, NumPy)
- SQL (Window Functions, CTEs, Joins, Aggregation, Query Optimization)

Data Engineering & Pipelines

- ETL / ELT
- Data Pipeline Design
- Batch Processing & Real-Time Stream Processing
- Data Integration
- Apache Spark (PySpark)
- Apache Airflow (Workflow Orchestration)
- dbt (SQL Transformation & Data Modeling)

Cloud & Data Platforms

- Microsoft Fabric (Lakehouse, OneLake, Data Factory, Notebooks, Power BI)
- Azure (Azure Data Factory, Azure Data Lake Storage Gen2, Azure Synapse Analytics)
- AWS (S3, EC2, IAM, Lambda, Glue)
- Databricks (Delta Lake, Lakehouse, Notebooks, Workflows)
- Snowflake (Data Warehousing, Virtual Warehouses, Stages, Snowpipe)

Data Modeling & Databases

- Data Modeling and Data Warehousing
- Schema Design (Star Schema, Snowflake Schema)
- ClickHouse, PostgreSQL, MySQL, SQL Server

Platform, Containers & Infrastructure

- Linux
- Docker & Docker Compose
- Kubernetes (Intermediate)
- Terraform (Familiar)
- YAML

Version Control

- Git & GitHub

Streaming & Messaging

- Apache Kafka (Topics, Partitions, Producers, Consumers, Consumer Groups, Offsets, Kafka Connect, Kafka Streams, ...)

CI / CD

- GitHub Actions
- Jenkins
- AWS CodePipeline, AWS CodeBuild

Monitoring & Observability

- Prometheus (Metrics collection fundamentals)
- Grafana (Dashboarding & visualization)

Soft Skills

- Team-oriented with strong collaboration skills
- Self-driven and capable of working independently
- Comfortable working in Agile Scrum teams.
- Strong passion for continuous learning and new technologies
- Problem-solving mindset with strong analytical thinking
- Highly organized, reliable, and committed to delivering quality work

Selected Data Engineering Projects (Hands-on, Independent)

- **Project: End-to-End Modern Data Engineering Pipeline (Neon PostgreSQL, Fivetran, Snowflake, dbt, and Databricks)**
 - Designed and implemented an end-to-end ELT pipeline using Neon PostgreSQL, Fivetran, Snowflake, dbt, and Databricks.

- Developed analytics-ready dimensional models using dbt, including staging, fact, dimension, testing, documentation. and business analytics layers
- Automated data ingestion from PostgreSQL into Snowflake using Fivetran.
- Developed PySpark-based RFM customer segmentation and created interactive Databricks SQL dashboards.
- **Project: End-to-End Microsoft Fabric Retail Lakehouse (OneLake, Data Factory, Lakehouse, PySpark & Power BI)**
 - Built an end-to-end retail analytics solution using Microsoft Fabric, OneLake, Data Factory, Lakehouse, PySpark, and Power BI.
 - Implemented a Medallion Architecture (Bronze, Silver, Gold) to ingest, transform, clean, and prepare retail data using PySpark and Delta tables.
 - Designed an analytics-ready Star Schema with fact and dimension tables in the Gold layer for downstream reporting.
 - Automated data orchestration using Microsoft Fabric Data Pipelines.
 - Built a semantic model and DAX measures for key retail KPIs, including orders, customers, basket size, and reorder behavior.
 - Developed interactive Power BI reports for executive overview, customer behavior, product performance, and reorder analysis.
- **Project: Azure End-to-End Data Engineering Pipeline (ADF, ADLS, Databricks, dbt & Synapse)**
 - Designed a cloud-native data pipeline using Azure Data Factory, ADLS, Azure Databricks, dbt, and Azure Synapse Analytics.
 - Built scalable ETL workflows to ingest, transform, and model e-commerce data using PySpark and the Medallion Architecture (Bronze, Silver, Gold).
 - Developed analytics-ready data models with dbt, including testing, documentation, and lineage.
 - Delivered interactive Power BI dashboards to visualize business KPIs and customer analytics.
- **Project: Data-Driven Career Recommendation Platform (Python, Streamlit & Machine Learning)**
 - Designed and developed an AI-powered career recommendation platform using Python and Streamlit.
 - Built a hybrid recommendation engine combining TF-IDF vectorization, Cosine Similarity, exact skill matching, and rule-based scoring to generate personalized job recommendations.
 - Implemented skill gap analysis, and personalized learning roadmaps using candidate profiles and market data.
 - Developed an interactive dashboard with visualizations, including recommendation insights.
- **Project: Energy Sensor Streaming Platform – Kafka Core**
 - Designed a Kafka-based streaming architecture for simulated energy sensor data.
 - Implemented Python producers and consumers for real-time ingestion.
 - Defined topic strategy, partitions, and retention for scalability and reliability.
 - Validated end-to-end data flow in a containerized Kafka environment.
- **Project: Kafka to ClickHouse Streaming Pipeline**
 - Built a real-time Kafka-to-ClickHouse ingestion pipeline using native ClickHouse Kafka Engine.
 - Implemented stream validation with ksqlDB and analytical storage using Materialized Views.
 - Enabled low-latency analytics on high-volume streaming data without Kafka Connect.
- **Project: Real-Time Stream Processing with ksqlDB**
 - Built SQL-based real-time stream processing using ksqlDB on Kafka.
 - Implemented event-time windowed aggregations and stateful tables.
 - Developed rule-based alert detection using continuous SQL queries.
- **Project: Real-Time Energy Alerting with Kafka and ksqlDB**
 - Designed an end-to-end real-time alerting pipeline on Kafka using ksqlDB.
 - Modeled streams and tables to process raw sensor data and generate alerts.
 - Implemented rule-based anomaly detection for power usage and temperature thresholds.
 - Built alert enrichment logic (type and severity) using SQL-based transformations.
 - Applied tumbling window aggregations to reduce alert noise and detect sustained anomalies.

EDUCATION AND QUALIFICATIONS

- Bachelor's degree, Computer Engineering | Software
- Master's degree, Industrial Management | Operation and Production Management
- Doctor of Philosophy – PhD, Technology Management | Innovation Management
- Digital Shaper Certificate – TechLabs (Data Science Track, 2026)
- Machine Learning — IMT | Certificate of Completion | 2023
- Prøve i Dansk 2 (PD2) – Danish Language Certificate, B1 Level

LANGUAGES

- Persian: Native
- English: Working Proficiency
- Danish: PD2 Level